

- 数据库实验报告 第6周
 - 实验内容：

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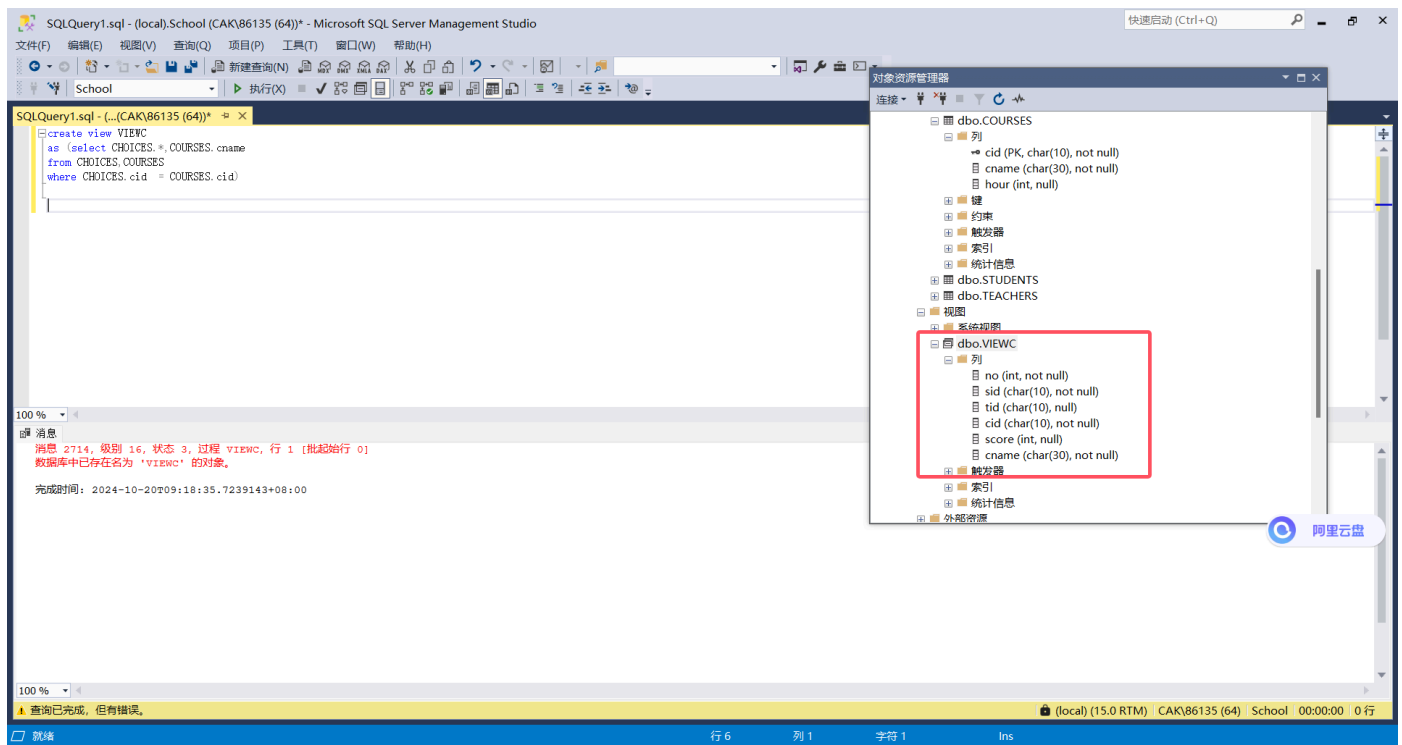
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实验内容：

(1)

创建VIEWWC语句如下：

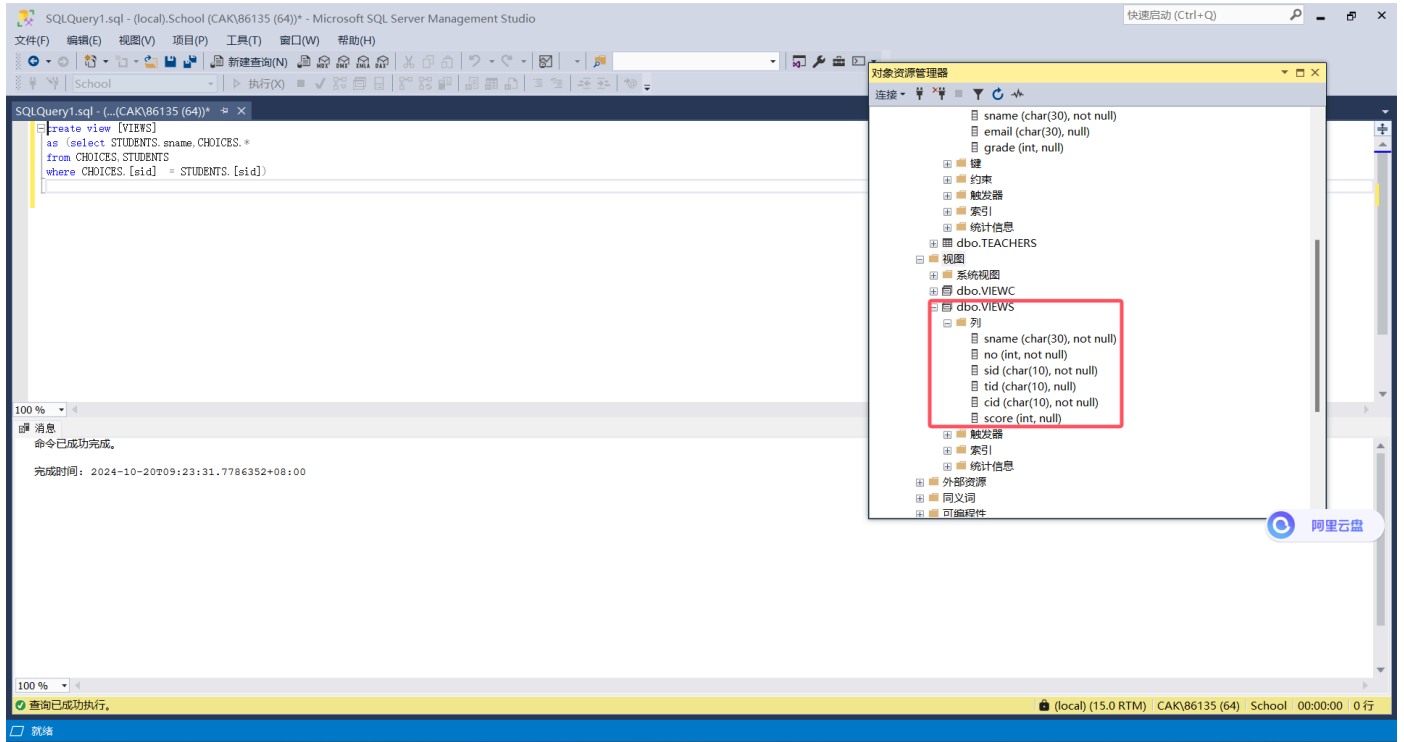
```
create view VIEWC
as (select CHOICES.*,COURSES.cname
from CHOICES,COURSES
where CHOICES.cid = COURSES.cid)
```



(2)

创建VIEWS语句如下：

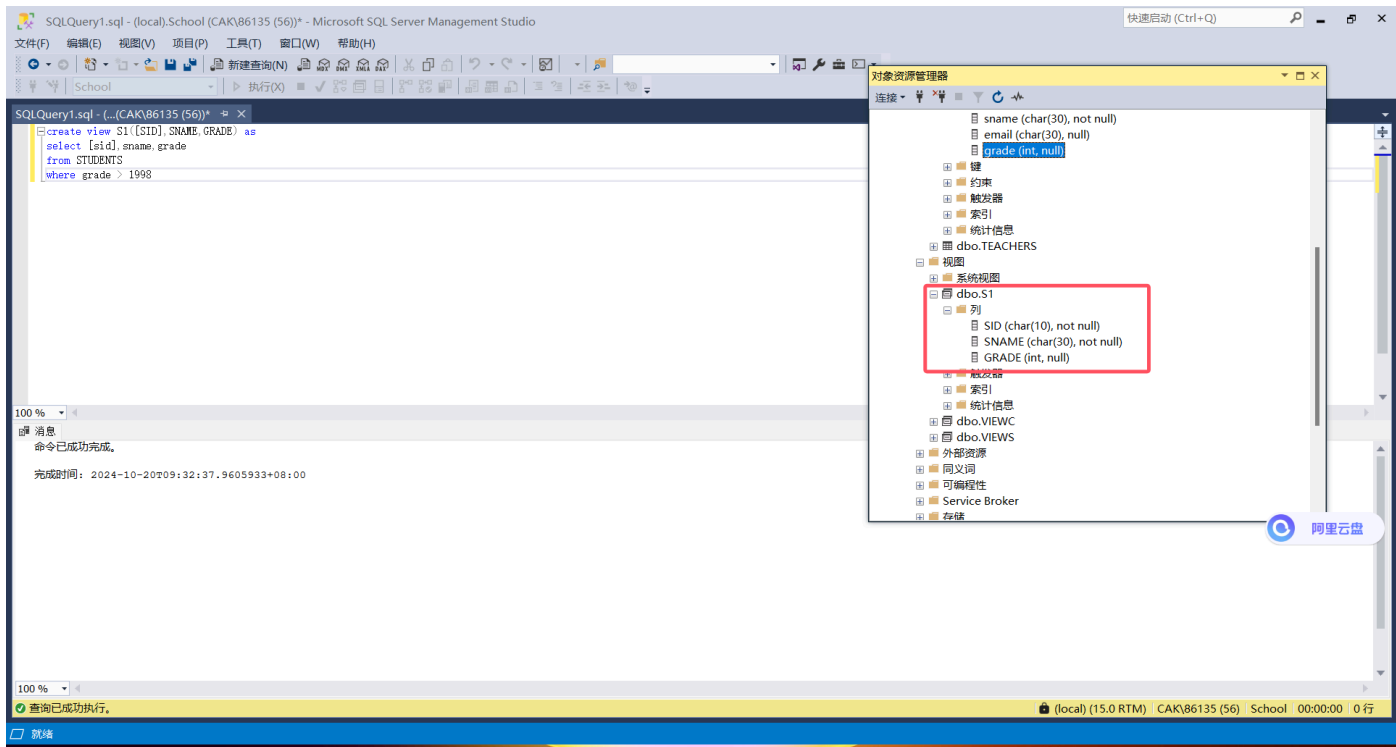
```
create view [VIEWS]
as (select STUDENTS.sname,CHOICES.*
from CHOICES,STUDENTS
where CHOICES.[sid] = STUDENTS.[sid])
```



(3)

创建视图S1语句如下，默认年份大于1998为符合条件的项：

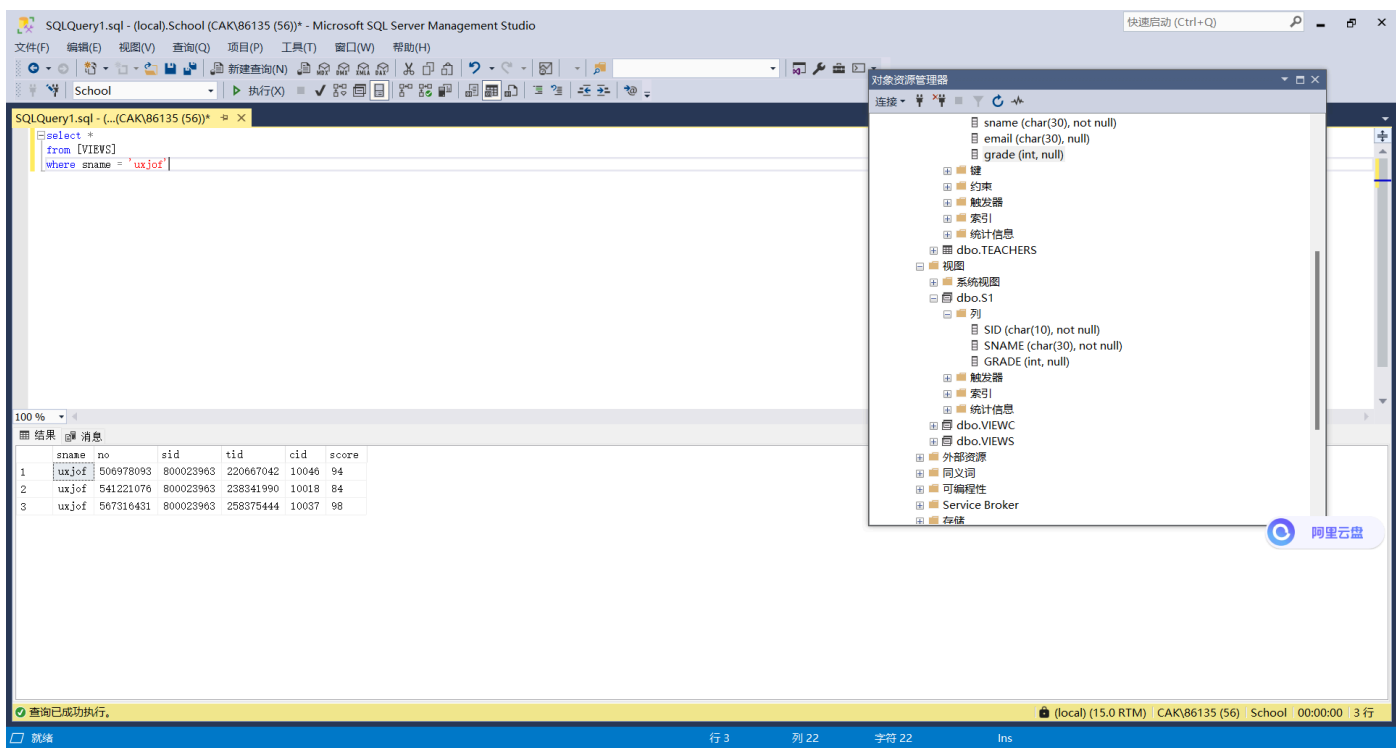
```
create view S1([SID],SNAME,GRADE) as
select [sid],sname,grade
from STUDENTS
where grade > 1998
```



(4)

通过VIEWS视图进行查询：

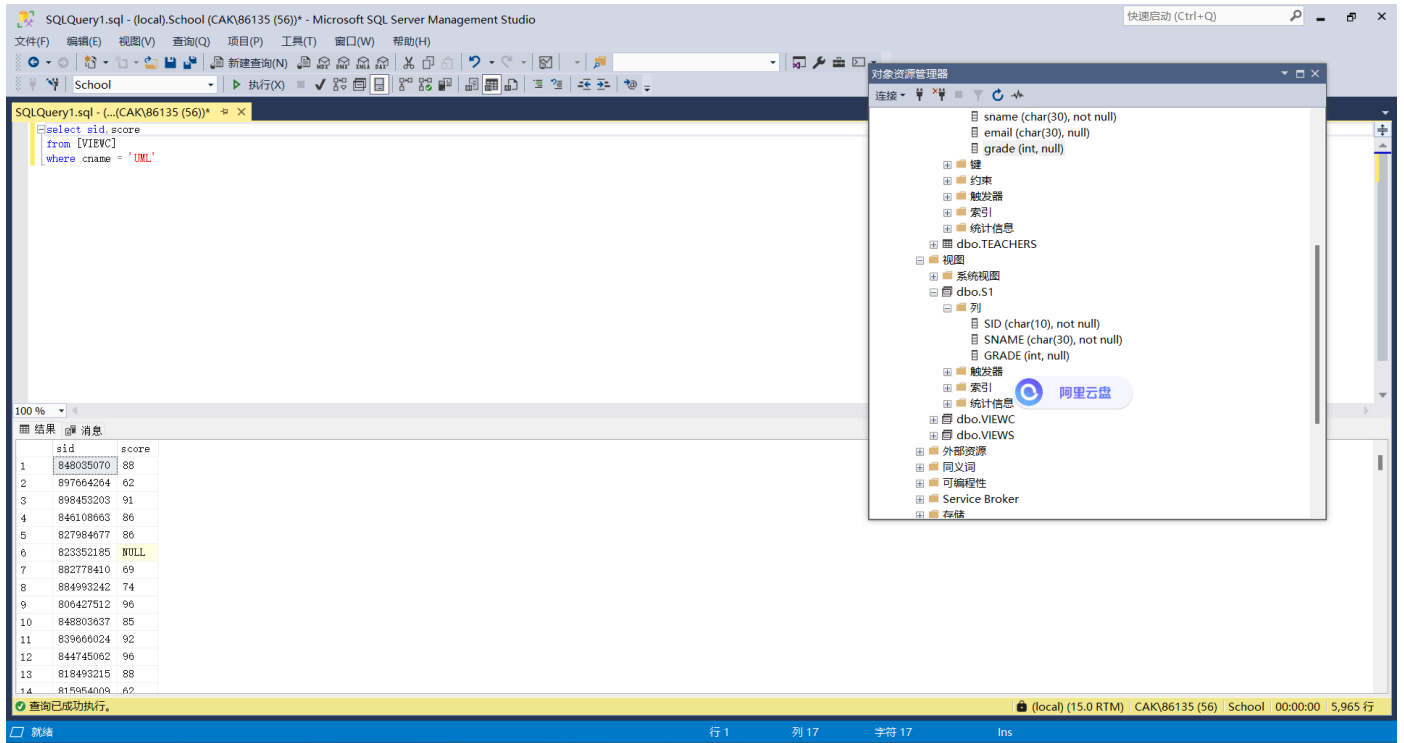
```
select *
from [VIEWS]
where sname = 'uxjof'
```



(5)

通过VIEWC进行查询：

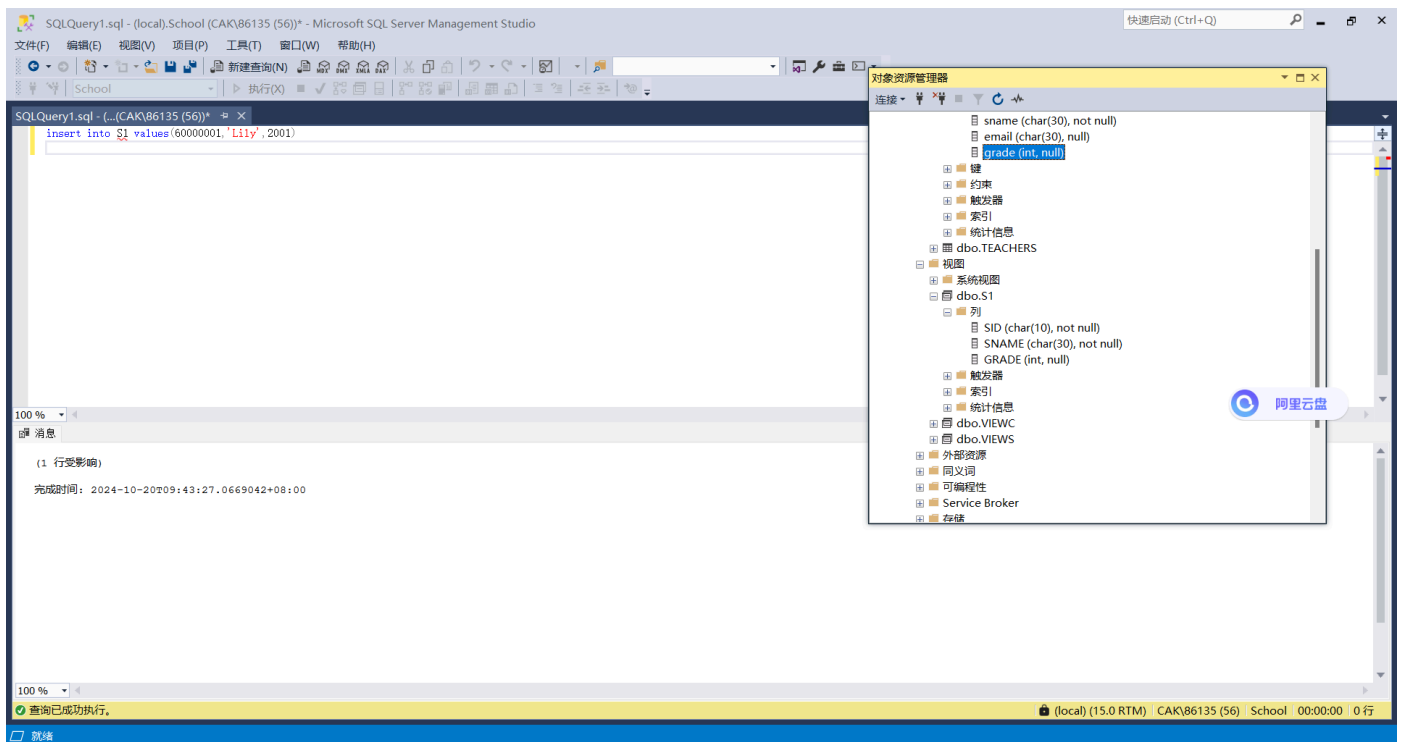
```
select sid,score
from [VIEWC]
where cname = 'UML'
```



(6)

视图插入语句如下：

```
insert into S1 values(60000001,'Lily',2001)
```



查询STUDENTS表，会发现STUDENTS表中也插入了此项，说明对于基本表也会执行插入操作：

SQLQuery1.sql - (local).School (CAK\86135 (60))* - Microsoft SQL Server Ma

文件(F) 编辑(E) 视图(V) 查询(Q) 项目(P) 工具(T) 窗口(W) 帮助(H)

新建查询(N) MDX DMX XMLA DAX

School 执行(X)

SQLQuery1.sql - (...(CAK\86135 (60))*

```
select *
from STUDENTS
where sname = 'Lily'
```

100 %

结果 消息

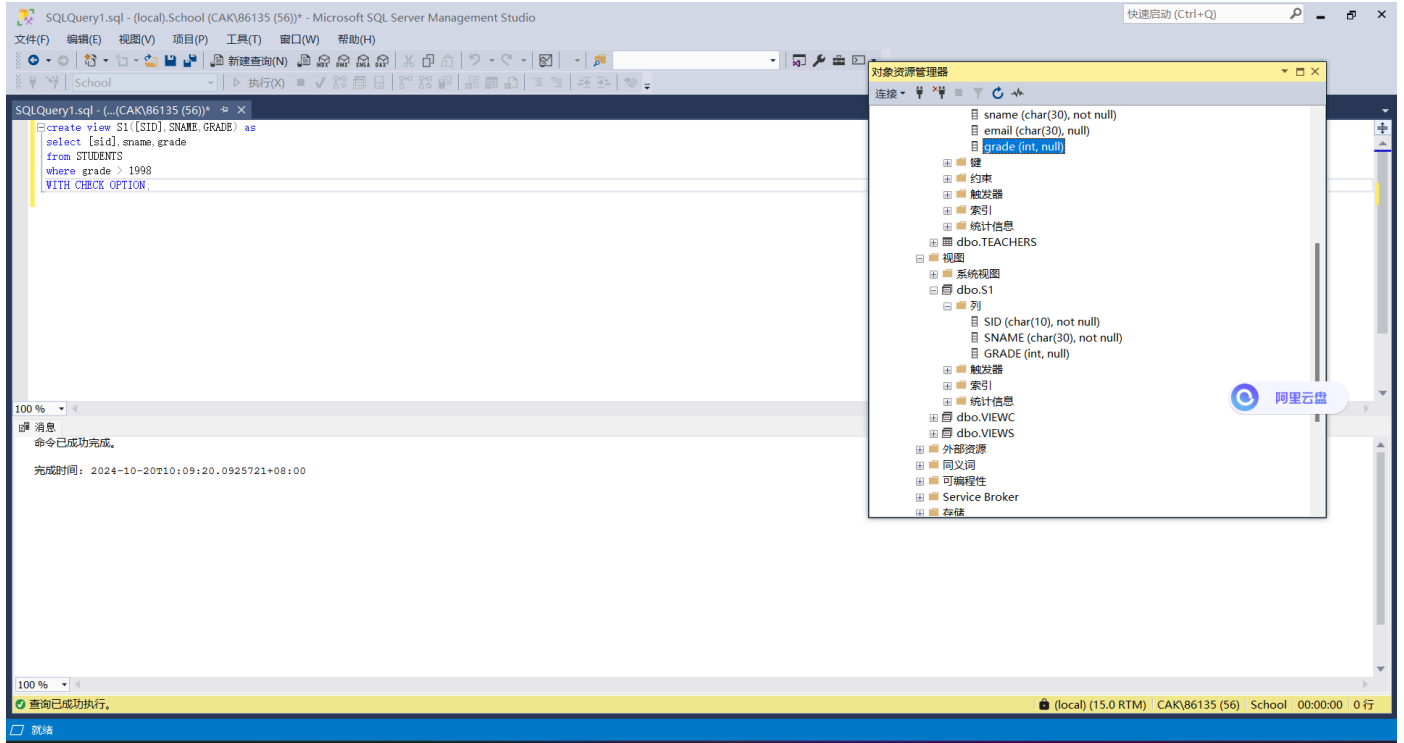
	sid	sname	email	grade
1	600000001	Lily	NULL	2001

(7)

通过如下语句创建包括更新和插入约束的视图S1：

```
create view S1([SID],SNAME,GRADE) as
select [sid],sname,grade
from STUDENTS
```

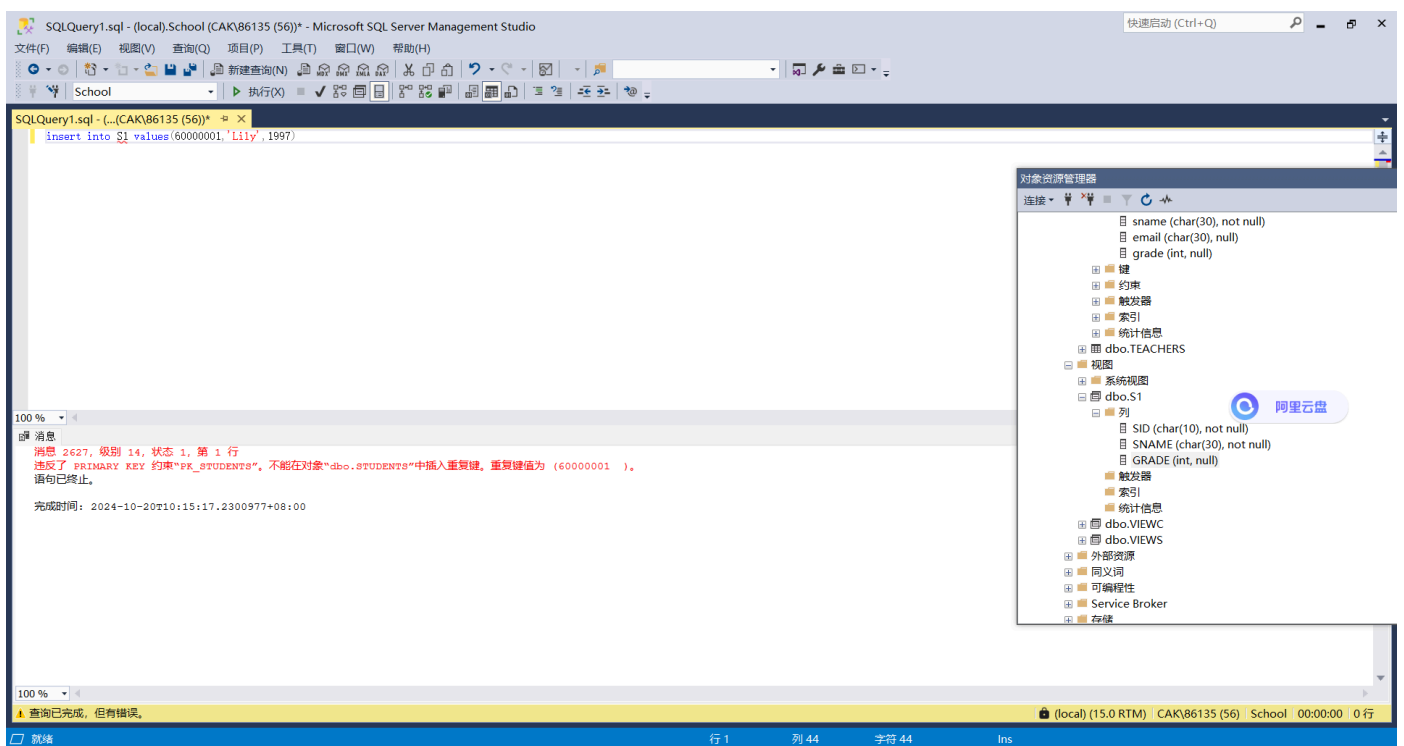
```
where grade > 1998
WITH CHECK OPTION;
```



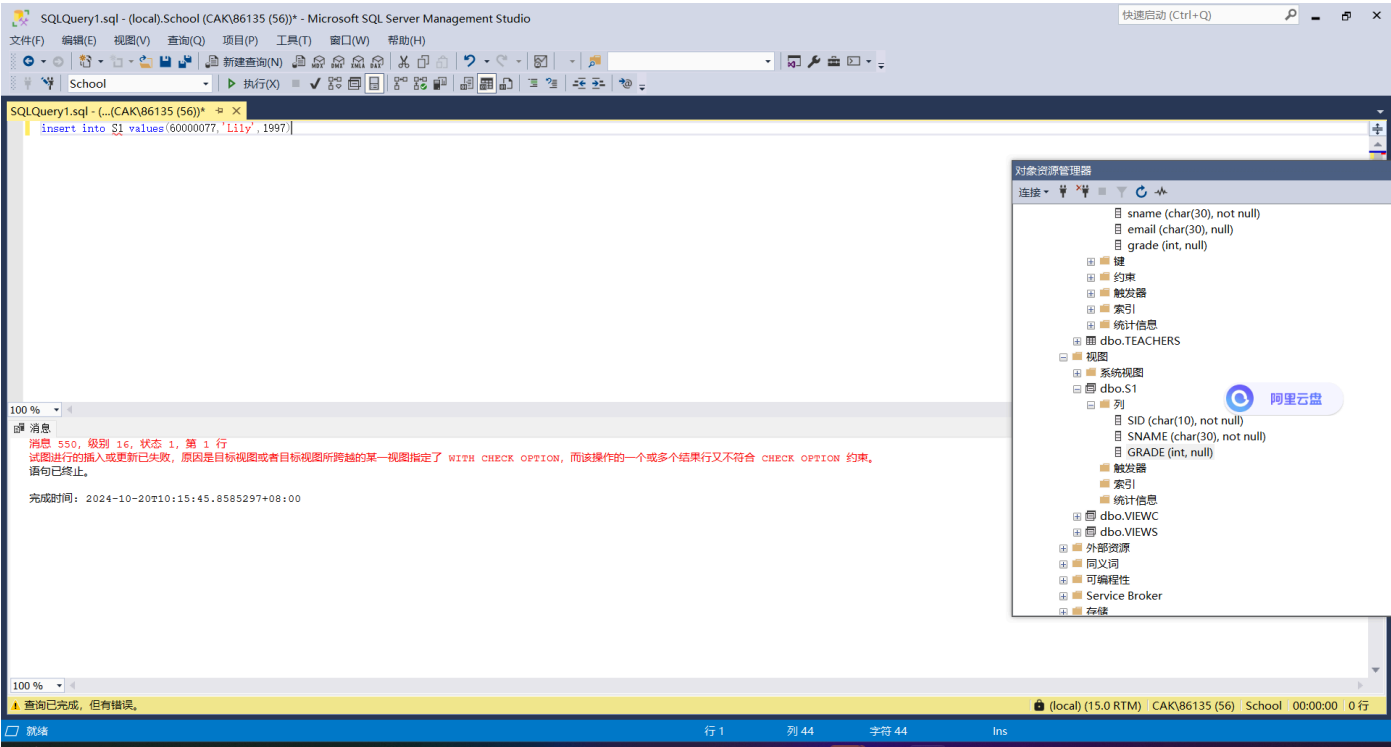
运行如下插入语句:

```
insert into S1 values(60000001,'Lily',1997)
```

会发现报错, 键值重复了:

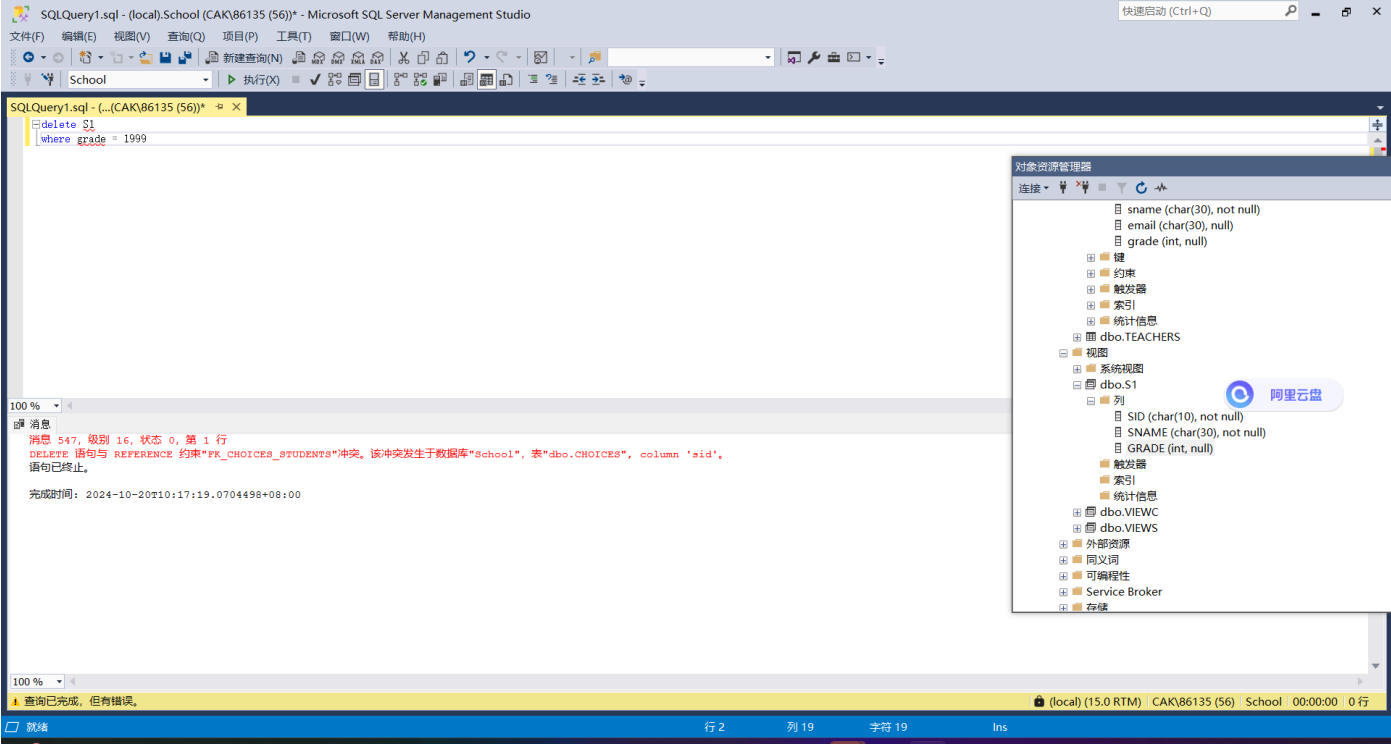


即使换一个合法的sid，也会报错，因为1997是比1998更早的年级，违反了>1998的条件：



运行如下语句，会发现由于外键约束的原因无法运行：

```
delete S1
where grade = 1999
```



这不是我们希望看到的，因此我首先取消对对应约束的检查，然后运行删除操作，再恢复约束检查，成功运行：

```
alter table CHOICES nocheck constraint FK_CHOICES_STUDENTS;  
delete S1  
where grade = 1999  
alter table CHOICES check constraint FK_CHOICES_STUDENTS;
```

SQLQuery1.sql - (...CAK\86135 (56))*

```
alter table CHOICES nocheck constraint FK_CHOICES_STUDENTS;  
delete S1  
where grade = 1999  
alter table CHOICES check constraint FK_CHOICES_STUDENTS;
```

100 %

消息

(6664 行受影响)

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更新和插入约束会保证操作符合约束条件，保证了数据库的安全性。

(8)

语句如下：

```
update [VIEWS]  
set score = score + 5  
where sname = 'uxjof';
```



```
update [VIEWS]
set score = score + 5
where sname = 'uxjof';
```

100 %

消息

(3 行受影响)

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(9)

删除视图操作如下:

```
drop view VIEWC;
drop view [VIEWS];
drop view S1;
```

